

COMBINATORICS

ASSIGNMENT 4

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1. How many paths of length k , P_k , are there in the complete graph on n edges, K_n ?

Solution. Let $V = \{v_0, \dots, v_{n-1}\}$ to be the vertices of a graph. Any ordered k -tuple chosen from V , like $\underbrace{(v_0, \dots, v_{i_0}, \dots, v_{i_1}, \dots, v_{i_{k-1}})}_{k\text{-tuples}}$ represents a path. For a specific set of indices $i_0, \dots, i_{k-1}$ we can represent an arbitrary path of length k as $(v_{i_0}, v_{i_1}, \dots, v_{i_{k-1}})$. Now consider the path $(v_{i_{k-1}}, \dots, v_{i_1}, v_{i_0})$ which is the same path as before but from the other direction. Thus, we see that we list every path twice when trying to count, once forwards and once backwards.

So we begin by choosing k vertices from our original n vertices in $\binom{n}{k}$ ways. Then, we choose our vertex ordering, which is a permutation in our index set, chosen in $k!$ ways, but like we said before we'll be double counting, so it turns out that we need $\frac{k!}{2}$. Thus, the total count follows as

$$\# = \binom{n}{k} \frac{k!}{2} = \frac{n!}{2(n-k)!}$$

□

2. How many 1-factors are there in K_{2n} ?

Solution. There are $(2n)!$ ways to arrange our $2n$ vertices. Let $V = \{v_1, \dots, v_{2n}\}$ be our vertex set, and let the indices $i_1, \dots, i_{2n}$ represent an arrangement $(v_{i_1}, v_{i_2}, \dots, v_{i_{2n}})$.

Now let's pair off the vertices in our arrangement in the following fashion $((v_{i_1}, v_{i_2}), (v_{i_3}, v_{i_4}), \dots, (v_{i_{2n-1}}, v_{i_{2n}}))$. Now we count this construction. There are $(2n)!$ ways to choose the indices on our vertices, but in our pairing as we've listed, the order of the n -pairs does not matter in the slightest so we will need to divide by $n!$ in the end. The ordering of the pair of vertices does not matter which happens in 2^n ways. Thus our count comes out as

$$\# = \frac{(2n)!}{2^n n!}.$$

□

3. How many 1-factors are there in the complete bipartite graph on n -vertices, $K_{n,n}$? Next show that it is D_n , the number of derangements.

Solution. We begin by setting $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$ to be the two distinct parts of $K_{n,n}$. A 1-factor is analogous to a perfect matching between x_i and y_i for all i noting that we can only pair a vertex once. For our count, we must remove a 1-factor. Because, we arbitrarily selected everything we can simply remove the 1-factor

$$M_0 = \{x_1y_1, \dots, x_ny_n\}.$$

So we are concerned with the graph $G = S_{n,n} \setminus M_0$, and we want to count the 1-factors of G . Furthermore, notice that every perfect matching x_i to y_i corresponds to a permutation in S_n let that be σ , except with the edge x_iy_i removed, giving us $\sigma(i) \neq i$.

We now count by inclusion/exclusion. There are $n!$ permutations of $[n]$. For each $i \in [n]$, let A_i be the set of permutations with $\sigma(i) = i$. Then consider $S \setminus (A_1 \cup \dots \cup A_n)$ via inclusion/exclusion

$$|S \setminus (A_1 \cup \dots \cup A_n)| = n! - \binom{n}{1}(n-1)! + \binom{n}{2}(n-2)! + \dots + (-1)^n \binom{n}{n}(n-n)!$$

Notice that $\binom{n}{k}(n-k)! = \frac{n!}{k!}$. So we have that

$$|S \setminus (A_1 \cup \dots \cup A_n)| = n! - \frac{n!}{1!} + \frac{n!}{2!} + \dots + (-1)^n \frac{n!}{n!}.$$

So we have that

$$\# \text{ of } K_{n,n} \setminus M_0 = \sum_{k=0}^n (-1)^k \frac{n!}{k!} (n-k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}.$$

Proof that D_n is the same count. We go back to our previous setup with $\{x_1, \dots, x_n\}$ and $\{y_1, \dots, y_n\}$ being our bipartition, and $M_0 = \{x_1y_1, \dots, x_ny_n\}$ being our removed 1-factor. Also notice that a 1-factor over $K_{n,n} \setminus M_0$ is equivalent to a permutation σ on $[n]$ with $\sigma(i) \neq i$ for any $i \in [n]$. Thus, σ is a derangement. Conversely, if σ is a derangement, then the set of edges $F_\sigma = \{x_iy_{\sigma(i)} : 1 \leq i \leq n\}$ is a 1-factor of $S_{n,n} \setminus M_0$ because $\sigma(i) \neq i$ we know that none of the edges is the edge $x_i - y_i$. So we have a bijection

$$D_n \leftrightarrow \{F_{n,n} \setminus M_0 \text{ 1-factors}\}.$$

So, from our last problem

$$|D_n| = |\{f_{n,n} \setminus M_0 \text{ 1-factors}\}| = \sum_{k=0}^n (-1)^k \frac{n!}{k!} (n-k)! = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

□

and we are done.